Portfolio Risk Modeling: Nvidia (NVDA) in the AI Market Boom

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1. Executive Summary

This report analyzes the daily risk of NVIDIA Corporation (NVDA) stock from January 1, 2023 to March 1, 2026. Using historical price data, daily returns were calculated and potential losses quantified through historical Value-at-Risk (VaR), Monte Carlo VaR, and Expected Shortfall (ES). Rolling 60-day VaR captures dynamic changes in risk over time. Results indicate that NVDA exhibits moderate daily volatility, with rare but significant losses in the worst 5% of trading days. The accompanying visualizations and metrics provide actionable insights for portfolio risk management.

The objective of this project is to measure NVDA's daily risk and quantify extreme potential losses. This analysis allows investors and portfolio managers to understand both typical day-to-day fluctuations and rare extreme losses.

2. Company Overview

NVIDIA Corporation (NVDA) is a leading technology company based in the United States, primarily known for designing graphics processing units (GPUs) and system-on-chip units for gaming, data centers, AI, and professional visualization. NVIDIA's stock is part of the NASDAQ 100 and is widely followed by investors due to its high growth potential and market volatility.

- Industry: Semiconductors / Technology
- Headquarters: Santa Clara, California, USA
- Founded: 1993
- Key products: GeForce GPUs, AI chips, data center solutions

Relevance to Risk Analysis:

- NVDA's stock price can be highly volatile because it is influenced by technology trends, product launches, and market sentiment.
- Its growth orientation and exposure to rapidly changing markets make daily returns more variable, which is why Value-at-Risk and Expected Shortfall metrics are important for investors and risk managers.

3. Business News and Market Context (2026)

In 2026, Nvidia's stock has been moving a lot in response to several key factors. First, the rapid growth of AI has pushed demand for Nvidia's GPUs, which are critical for AI workloads in data centers and cloud computing. This has helped the stock rise, but also made it more sensitive to any news.

At the same time, earnings reports have been mixed. While the company continues to grow, some quarterly guidance has been conservative. Investors react quickly to this combination of strong results and cautious outlooks, which leads to short-term swings in the stock price.

Regulatory speculation has also played a role. Concerns about export controls or market restrictions in certain regions sometimes create sudden drops or spikes, even when the company's fundamentals are solid.

As a result, NVDA has been experiencing high volatility, meaning the stock can move up or down significantly from day to day. For investors, this means:

- Short-term trading is risky. Daily swings can be large, so tools like Value-at-Risk (VaR) and Expected Shortfall (ES) help estimate how much could be lost on a bad day.

- Portfolio decisions need care. Holding too much NVDA in a portfolio can increase overall risk. Monitoring rolling VaR helps spot periods when risk is higher.

- Risk management matters. Combining these metrics with awareness of news gives investors a better chance of staying on top of potential losses without overreacting to normal market noise.

Nvidia's stock offers strong growth potential, but it can swing sharply in response to AI trends, earnings news, or regulatory events. Quantitative risk measures like VaR and ES give investors a clearer picture of possible losses and help them make smarter decisions.

4. Business Problem

Problem Statement:
Investors in Nvidia stock face significant downside risk due to AI sector volatility. How much could an investor potentially lose over a 1-day or 10-day horizon at a 95% confidence level?

Objectives:

1. Calculate Historical VaR for Nvidia stock based on past returns.

2. Simulate Monte Carlo VaR to estimate potential losses under future uncertainty.

3. Compare the two methods and visualize the risk distribution.
4. Provide actionable insights for portfolio management.

5. Data Description

- **Data Source**: Yahoo Finance via yfinance library.
- **Ticker**: NVDA (Nvidia Corporation)
- **Timeframe**: Last 3 years of daily stock prices (2023–2026).
- **Variables**: Daily adjusted close prices → calculate daily returns.
- **Preprocessing**: Remove missing values, compute percentage returns.

6. Methodology

Step 1: Calculate Daily Returns

- Convert adjusted close prices into **percentage returns**:

$$\text{Return}_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Step 2: Historical VaR

- Use **past returns** to calculate the **5th percentile** for 95% confidence VaR.
- Formula:

$$\text{VaR}_{\text{Historical}} = \text{Quantile}(\text{Returns}, 0.05)$$

Step 3: Monte Carlo VaR

- Simulate **10,000+ future returns** assuming returns follow a normal distribution with mean = historical mean, standard deviation = historical std.
- Compute the 5th percentile of simulated returns.

Step 4: Visualization

- Histogram of historical returns with VaR lines.
- Overlay Monte Carlo simulated distribution.
- Optional: cumulative distribution function (CDF) plot.

Step 5: Insights

- Compare Historical vs Monte Carlo VaR: which method is more conservative?
- Discuss potential implications for an investor (e.g., exposure limits, stop-loss planning).

7. Python Implementation Outline

NVDA

```
import yfinance as yf
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# ------------------------------
# 1. Download Data
# ------------------------------
nvda = yf.download('NVDA', start='2023-01-01', end='2026-03-01', auto_adjust=True)

# ------------------------------
# 2. Compute Daily Returns
# ------------------------------
nvda['Return'] = nvda['Close'].pct_change().dropna()

# ------------------------------
# 3. Descriptive Statistics
# ------------------------------
mean_return = nvda['Return'].mean()
std_return = nvda['Return'].std()
min_return = nvda['Return'].min()
max_return = nvda['Return'].max()

print("Descriptive Statistics:")
print(f"Mean Daily Return: {mean_return:.4%}")
print(f"Standard Deviation: {std_return:.4%}")
```

```
print(f"Minimum Daily Return: {min_return:.2%}")
print(f"Maximum Daily Return: {max_return:.2%}")

# ------------------------------
# 4. Value-at-Risk (VaR)
# ------------------------------
# Historical VaR (5th percentile)
VaR_hist = nvda['Return'].quantile(0.05)

# Monte Carlo Simulation VaR
mu = mean_return
sigma = std_return
simulations = np.random.normal(mu, sigma, 100000)
VaR_mc = np.percentile(simulations, 5)

# Expected Shortfall (Historical)
ES_hist = nvda['Return'][nvda['Return'] <= VaR_hist].mean()

print("\nValue-at-Risk (95% confidence):")
print(f"Historical VaR: {VaR_hist:.2%}")
print(f"Monte Carlo VaR: {VaR_mc:.2%}")
print(f"Historical Expected Shortfall (ES): {ES_hist:.2%}")

# ------------------------------
# 5. Rolling 60-Day Historical VaR
# ------------------------------
nvda['VaR_rolling_60'] = nvda['Return'].rolling(window=60).quantile(0.05)

# ------------------------------
# 6. Visualization
# ------------------------------

fig, axes = plt.subplots(3, 1, figsize=(12,14), gridspec_kw={'height_ratios':[1,2,2]})

# --- Plot 1: Closing Price over Time ---
axes[0].plot(nvda.index, nvda['Close'], color='blue')
axes[0].set_title('NVDA Adjusted Close Price (2023-2026)')
axes[0].set_ylabel('Price ($)')
axes[0].grid(True)

# --- Plot 2: Rolling 60-Day VaR ---
axes[1].plot(nvda.index, nvda['VaR_rolling_60'], color='r', label='Rolling 60-Day 95% VaR')
axes[1].axhline(VaR_hist, color='g', linestyle='dashed', label=f'Overall Historical VaR:
{VaR_hist:.2%}')
axes[1].set_title('NVDA Rolling 95% Historical VaR Over Time')
axes[1].set_ylabel('Daily Return (VaR)')
axes[1].legend()
axes[1].grid(True)
```

```
# --- Plot 3: Histogram with Shaded 5% Worst Returns ---
counts, bins, patches = axes[2].hist(nvda['Return'], bins=50, alpha=0.6, label='Daily Returns',
color='skyblue')
for patch, bin_edge in zip(patches, bins):
    if bin_edge <= VaR_hist:
        patch.set_facecolor('salmon')  # Shade worst 5% returns

# VaR Lines
axes[2].axvline(VaR_hist, color='r', linestyle='dashed', linewidth=2, label=f'Historical VaR:
{VaR_hist:.2%}')
axes[2].axvline(VaR_mc, color='g', linestyle='dashed', linewidth=2, label=f'Monte Carlo VaR:
{VaR_mc:.2%}')
# ES line
axes[2].axvline(ES_hist, color='purple', linestyle='dotted', linewidth=2, label=f'Expected Shortfall
(ES): {ES_hist:.2%}')

# Annotate VaR and ES
axes[2].text(VaR_hist, axes[2].get_ylim()[1]*0.9, f'{VaR_hist:.2%}', color='r', rotation=90,
verticalalignment='center')
axes[2].text(VaR_mc, axes[2].get_ylim()[1]*0.8, f'{VaR_mc:.2%}', color='g', rotation=90,
verticalalignment='center')
axes[2].text(ES_hist, axes[2].get_ylim()[1]*0.7, f'{ES_hist:.2%}', color='purple', rotation=90,
verticalalignment='center')

axes[2].set_title('NVDA Daily Returns Distribution with 95% VaR & Expected Shortfall')
axes[2].set_xlabel('Daily Return')
axes[2].set_ylabel('Frequency')
axes[2].legend()
axes[2].grid(True)

plt.tight_layout()
plt.show()

# ------------------------------
# 7. Optional: Rolling VaR Table Snapshot
# ------------------------------
print("\nSample Rolling 60-Day VaR (last 10 entries):")
print(nvda[['VaR_rolling_60']].tail(10))
```

Apple comparison

```
import yfinance as yf
import numpy as np
import pandas as pd
```

```
# Download Apple data
aapl = yf.download('AAPL', start='2023-01-01', end='2026-03-01', auto_adjust=True)

# Compute daily returns
aapl['Return'] = aapl['Close'].pct_change().dropna()

# Descriptive stats
aapl_mean = aapl['Return'].mean()
aapl_std = aapl['Return'].std()
aapl_min = aapl['Return'].min()
aapl_max = aapl['Return'].max()

# VaR & ES
aapl_VaR_hist = aapl['Return'].quantile(0.05)
simulations = np.random.normal(aapl_mean, aapl_std, 100000)
aapl_VaR_mc = np.percentile(simulations, 5)
aapl_ES_hist = aapl['Return'][aapl['Return'] <= aapl_VaR_hist].mean()

# Rolling 60-day VaR
aapl['VaR_rolling_60'] = aapl['Return'].rolling(window=60).quantile(0.05)

# Combine NVDA and AAPL for comparison table
comparison = pd.DataFrame({
    'Metric': ['Mean Return', 'Std Deviation', 'Min Return', 'Max Return', 'Historical VaR', 'Monte Carlo
VaR', 'Expected Shortfall (ES)'],
    'NVDA': [nvda['Return'].mean(), nvda['Return'].std(), nvda['Return'].min(), nvda['Return'].max(),
VaR_hist, VaR_mc, ES_hist],
    'AAPL': [aapl_mean, aapl_std, aapl_min, aapl_max, aapl_VaR_hist, aapl_VaR_mc, aapl_ES_hist]
})
print(comparison)
```

8. Visualizations and Results

Descriptive Statistics:
Mean Daily Return: 0.3672%
Standard Deviation: 3.1297%
Minimum Daily Return: -16.97%
Maximum Daily Return: 24.37%

Value-at-Risk (95% confidence):
Historical VaR: -4.14%
Monte Carlo VaR: -4.75%
Historical Expected Shortfall (ES): -6.20%
Sample Rolling 60-Day VaR (last 10 entries):
Price VaR_rolling_60

Ticker	
Date	
2026-02-13	-0.032736
2026-02-17	-0.032736
2026-02-18	-0.032736
2026-02-19	-0.032736
2026-02-20	-0.032736
2026-02-23	-0.032736
2026-02-24	-0.032736
2026-02-25	-0.032736
2026-02-26	-0.034305
2026-02-27	-0.038325

The analysis of NVDA's daily returns from January 2023 to March 2026 shows that the stock has moderate day-to-day volatility. On average, NVDA gained about 0.37% per day, but typical daily movements were around ±3.13%. While most days had small changes, there were some large swings, with the biggest loss at -16.97% and the biggest gain at 24.37%.

The 95% Value-at-Risk (VaR) measures potential daily losses under normal conditions. The historical VaR of -4.14% means that in roughly 1 out of 20 days, NVDA could lose more than 4.1%. The Monte Carlo VaR of -4.75% is slightly higher, showing that simulated scenarios predict slightly larger potential losses.

The Expected Shortfall (ES) of -6.20% gives the average loss if NVDA falls into the worst 5% of days. This shows that extreme losses can be larger than the VaR threshold.

Looking at the rolling 60-day VaR, the risk has mostly been between -3.27% and -3.83%, with a slight increase toward the end of February 2026. This indicates that daily risk can change over time, reflecting periods of higher volatility.

NVDA Event Timeline

Time Period	Stock Behavior	What Happened & Why
Early 2023	Modest uptrend	NVIDIA was still recovering after earlier market downturns. Demand for GPUs in gaming and data centers began strengthening.
Mid 2023	Steady rise	Early signs of AI investment growth. Companies started spending on AI infrastructure, increasing demand for NVIDIA GPUs.

Late 2023	Sharper gains	Analysts raised earnings forecasts as NVIDIA reported strong quarterly revenue — especially data center sales powering AI workloads.
Early 2024	Big price jump	*AI boom accelerates*: NVIDIA's data center segment reported huge growth, beating expectations. Investors became more confident about long-term demand for AI chips.
June 2024	Continued growth + stock split effect	NVIDIA announced a **10-for-1 stock split**, making shares easier for smaller investors to buy. This often boosts trading interest and can support higher prices.
Mid 2024	Upside but rising volatility	The market grew cautious about the high valuations: some profit-taking and rotation into other sectors. This appeared as *higher short-term risk* in the VaR chart.
Late 2024	Price keeps rising	Continued strong earnings from AI product sales — especially H100 and AI accelerators — kept investors optimistic.
Early 2025	Price dips + higher VaR (risk)	New product announcements (next-gen GPUs) were positive, *but guidance was conservative*. Some traders took profits on strong gains, increasing volatility.
Spring–Mid 2025	Sharp risk spike + temporary sell-off	Regulatory / export controls impacted NVIDIA's ability to sell some chips into China, raising short-term uncertainty about revenue. This showed up as higher potential daily losses (VaR spiking).
Late 2025	Price recovery and new highs	Easing of some export concerns + continued AI demand + major partnerships pushed the stock back up, often to new record levels.
Early 2026	High price with moderate risk	After large gains, the stock continued to trade at elevated levels. Risk normalized compared to earlier spikes, though short-term swings still occur.

Observations:

- Strong bullish trend: Price rises from ~$20 in early 2023 to ~$200 in early 2026, a roughly 10x increase.

- Growth is uneven: The chart shows periods of rapid ascent (early 2024, late 2025) interspersed with brief corrections.

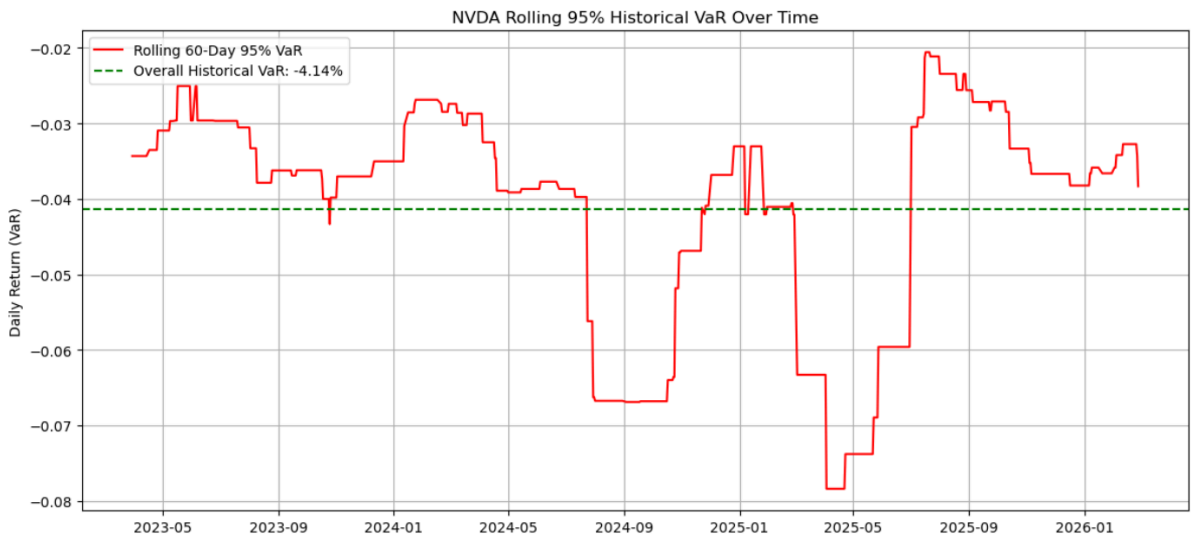
- Volatility: Small dips indicate short-term corrections, suggesting the market was occasionally pricing in risk or uncertainty.

Critical Analysis:

- Price movement reflects fundamentals: The surge corresponds to NVIDIA's dominance in AI GPUs and strong earnings from data center products. Growth was driven by real adoption of AI technologies in enterprise and cloud computing.

- Short-term corrections are healthy: These dips show the market responding to valuation concerns, profit-taking, and geopolitical/regulatory uncertainties, not a collapse of the underlying business.

- Pattern of acceleration: Sharp upward swings (especially early 2024 and late 2025) suggest that positive news (earnings surprises, stock split, AI hype) triggers exponential growth in investor confidence, common in high-growth tech stocks.



Observations:

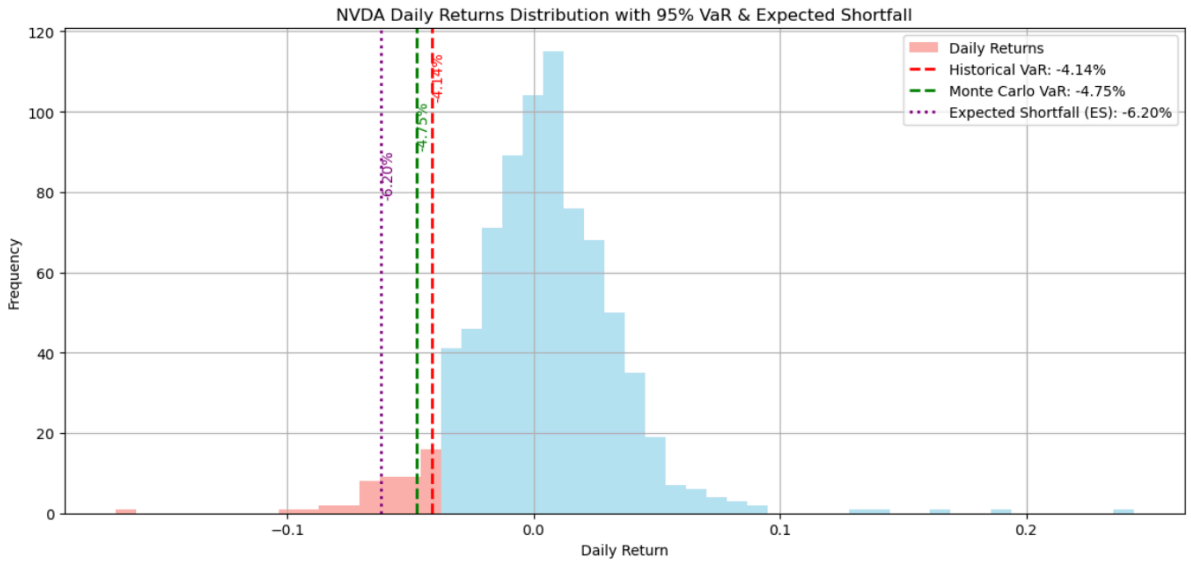
- The red line represents rolling 60-day 95% VaR, showing potential daily losses in extreme scenarios.

- Average VaR: ~–4.14% (green dashed line).

- Volatility spikes: Some periods (mid-2024, early and mid-2025) see VaR spike to –7% or –8%, indicating higher daily risk.

Critical Analysis:

- VaR spikes align with market events:
    - **Mid-2024:** Market digesting fast gains, profit-taking and cautious sentiment.
    - **Early 2025:** New product announcements with conservative guidance caused temporary uncertainty.
    - **Mid-2025:** Regulatory/export concerns (China restrictions on AI chips) increased short-term risk.
- VaR shows temporary vs. structural risk: While spikes are dramatic, they are short-term and event-driven, not signaling long-term deterioration. Long-term upward trend in the price chart validates this.
- Investor takeaway: The rolling VaR effectively highlights periods when risk management matters most, e.g., position sizing or hedging during high-risk periods.



Line	Color	What it means

Historical 95% VaR	Red dashed	The worst 5% of losses historically, a daily loss worse than –4.14% happens only 5% of the time.
Monte Carlo 95% VaR	Green dashed	Another estimate of the worst 5% loss using simulations, here about –4.75%, slightly more conservative.
Expected Shortfall (ES)	Purple dotted	The average loss if you do hit the worst 5% days, here –6.20%. This is bigger than VaR because it accounts for extreme losses beyond the 95% threshold.

The NVDA daily returns distribution graph shows how the stock's daily price changes are spread over time. Most daily returns cluster close to zero, indicating that on typical days, the stock experiences small gains or losses. However, the graph also highlights the potential for more extreme moves. The 95% historical VaR, shown by the red dashed line at –4.14%, indicates that on the worst 5% of days, NVDA could lose more than 4% in a single day. The Monte Carlo VaR (green dashed line, –4.75%) provides a slightly more conservative estimate based on simulated scenarios, suggesting that rare losses could be even larger. The Expected Shortfall (purple dotted line, –6.2%) goes further, showing the average loss when one of those worst-case days occurs, capturing the tail risk beyond the VaR threshold. In essence, the graph illustrates that while NVDA usually moves modestly day-to-day, it carries a small but real risk of sharp declines, and understanding this helps investors prepare for extreme market events without overreacting to normal fluctuations.

Comparative analysis

```
Metric       NVDA      AAPL
0              Mean Return  0.003672  0.001097
1            Std Deviation  0.031297  0.016199
2               Min Return -0.169682 -0.092456
3               Max Return  0.243696  0.153288
4           Historical VaR -0.041360 -0.024380
5          Monte Carlo VaR -0.047536 -0.025609
6  Expected Shortfall (ES) -0.061962 -0.036135
```

1. Daily Returns and Volatility:

- NVDA's mean daily return is ~0.37%, compared to ~0.11% for Apple, showing that NVDA has slightly higher average growth.

- NVDA is also more volatile, with a standard deviation of 3.13% versus Apple's 1.62%. This means NVDA's daily price movements are more pronounced.

- Extreme movements are larger for NVDA: its minimum daily return is -16.97% and maximum is 24.37%, while Apple's range is -9.25% to 15.33%.

2. Value-at-Risk (VaR):

- Historical 95% VaR indicates that on the worst 5% of trading days, NVDA could lose 4.14% in a single day, whereas Apple's worst 5% loss is smaller at 2.44%.

- Monte Carlo VaR provides slightly more conservative estimates: -4.75% for NVDA and -2.56% for Apple.

3. Expected Shortfall (ES):

- ES measures the average loss in the worst 5% of days. NVDA's ES is -6.20%, compared to -3.61% for Apple. This confirms that when extreme losses occur, NVDA's potential downside is significantly higher.

4. Interpretation:

- NVDA is a high-growth, high-volatility stock: larger swings, higher potential gains, but also higher downside risk.

- Apple is more stable, with smaller daily fluctuations and lower extreme losses.

- This comparison illustrates how risk measures like VaR and ES differ across companies, which is valuable for portfolio allocation and risk management. Investors holding NVDA may require hedging or position sizing strategies, while Apple may be used to balance portfolio stability.

9. Business Insights / Recommendations

- NVDA stock shows high downside risk, particularly during AI market corrections.

- Investors should:

    - Consider position sizing to limit exposure.

    - Use stop-loss strategies or hedging instruments.

    - Monitor sector volatility trends to adjust allocation.

NVDA offers significant growth potential, but its daily returns exhibit occasional sharp losses. Investors should balance exposure with diversification, employ stop-losses or hedges during periods of elevated risk, and monitor company and sector news. Rolling VaR and Expected Shortfall metrics provide a quantitative framework to anticipate and manage potential downside, enabling confident investment decisions even in a volatile market.

10. Conclusion

- The project demonstrates risk quantification for a high-volatility stock.
- Historical and Monte Carlo VaR provide complementary insights: historical reflects past experience, Monte Carlo simulates possible extreme future scenarios.
- Applying this methodology can inform investment decisions and portfolio risk management.

11. References / Data Sources

- Yahoo Finance (yfinance)
- Recent Nvidia news articles from Bloomberg, CNBC, Wall Street Journal
- Python Libraries: Pandas, NumPy, Matplotlib